

NAME

standard -- Add standard stars to sensitivity file

USAGE

standard input [records] output

PARAMETERS**input**

List of input standard star spectra or root names if using the record number extension format. All spectra of the same aperture must be of the same standard star. In beam switch mode or when the same star parameter is set all spectra must be of the same standard star regardless of aperture number. Normally the spectra will not be extinction corrected but if then the extinction file should also be given and the same extinction file should be used with **sensfunc**.

records(imred.irsand imred.iidsonly)

List of records or ranges of records to be appended to the input spectra names when using record number extension format. The syntax of this list is comma separated record numbers or ranges of record numbers. A range consists of two numbers separated by a . A null list may be used if no record number extensions are desired. This is a positional query parameter only if the record format is specified.

output

The name of a text file which will contain the output from **standard**. Each execution of **standard** appends to this file information about the standard stars, the calibration bandpasses, and observed counts (see the DESCRIPTION section for more details). The output must be explicitly deleted by the user if the filename is to be reused.

samestar= yes

Is the same star in all apertures? If set to no then each aperture may contain a different standard star. The standard star name is queried each time a new aperture is encountered. Note that this occurs only once per aperture and multiple spectra with the same aperture number must be of the same star. If set to yes the standard star name is only queried once. When in beam switch mode this parameter is ignored since all apertures must contain the same star.

beam_switch= no

Beam switch the spectra? If yes then a beam switch mode is used for the spectra in which successive pairs of object and sky observations from the same aperture are sky subtracted. This requires that the object type flag OFLAG be present and that the spectra are appropriately ordered. All object observations must be of the same standard star and the *samestar* parameter is ignored.

apertures =""

List of apertures to be selected from the input list of spectra. If is specified then all apertures are selected. The syntax is the same as the record number extensions.

bandwidth= INDEF, bandsep= INDEF

Bandpass widths and separations in wavelength units. If INDEF then the default bandpasses are those given in the standard star calibration file. If values for these parameters are specified then a of bandpasses of equal width and separation are defined over the range of the input spectrum. In both cases the default bandpasses can be changed interactively if desired.

fnuzero= 3.68e-20

The absolute flux per unit frequency at an AB magnitude of zero. This is to convert the calibration AB magnitudes to absolute flux by the formula

$$f_{\nu} = \text{fnuzero} * 10. ** (-0.4 * m_{\text{AB}})$$

The flux units are also determined by this parameter. However, the frequency to wavelength interval conversion assumes frequency in hertz. The default value is based on a calibration of Vega at 5556 Angstroms of $3.52\text{e-}20$ ergs/cm²/s/Hz for an AB magnitude of 0.0336. This default value is that used in earlier versions of this task which did not allow the user to change this calibration.

extinction= <no default>

Extinction file used to make second order extinction corrections across the bandpasses. The default value is redirected to the package parameter of the same name. See **lcalib** for a extinctionfiles. Normally the input spectra will not be extinction corrected. But if they are this file will be used to remove the extinction and then the same file should be specified in **sensfunc**. Note that one can choose to use a null extinction file in both.

caldir =")_.caldir"

Calibration directory containing standard star data. The default value of ")_.caldir" means to use the package parameter "caldir". A list of standard calibration directories may be obtained by listing the file "onedstds\$README"; for example:

```
cl> page onedstds$README
```

The user may copy or create their own calibration files and specify the directory. The directory "" refers to the current working directory. The standard calibration directory for blackbody curves is "onedstds\$blackbody/".

observatory =")_.observatory"

Observatory at which the spectra were obtained if not specified in the image header by the keyword OBSERVAT. The default is a redirection to look in the parameters for the parent package for a be one of the observatories in the observatory database, "observatory" to select the observatory defined by the environment variable "observatory" or the parameter **observatory.observatory**, or "obsvars" to select the current parameters set in the **observatory** task. See help for **observatory** for additional information.

interact= no

If set to no, then the default wavelength set (either that from the star calibration file or the set given by the *bandwidth* and *bandsep* parameters) is used to select wavelength points along the spectrum where the sensitivity is measured. If set to yes, the spectra may be plotted and the bandpasses adjusted.

graphics ="stdgraph"

Graphics output device for use with the interactive mode. Normally this is the user's graphics terminal.

cursor =""

Graphics cursor input for use with the interactive mode. When null the standard graphics cursor is used otherwise the specified file is used.

star_name

The name of the star observed in the current series of spectra. Calibration data for the star must be in the specified calibration directory "caldir". This is normally a interactive query parameter and

should not be specified on the command line unless all spectra are of the same standard star.

The following three queried parameters apply if the selected calibration file is for a blackbody.

mag The magnitude of the observed star in the band given by the *magband* parameter. If the magnitude is not in the same band as the blackbody calibration file then the magnitude may be converted to the calibration band provided the "params.dat" file containing relative magnitudes between the two bands is in the calibration directory

magband

The standard band name for the input magnitude. This should generally be the same band as the blackbody calibration file. If it is not the magnitude will be converted to the calibration band.

teff The effective temperature (deg K) or the spectral type of the star being calibrated. If a spectral type is specified a "params.dat" file must exist in the calibration directory. The spectral types are specified in the same form as in the "params.dat" file. For the standard blackbody calibration directory the spectral types are specified as A0I, A0III, or A0V, where A can be any letter OBAFGKM, the single digit subclass is between 0 and 9, and the luminosity class is one of I, III, or V. If no luminosity class is given it defaults to dwarf.

The following two parameters are queried if the image does not contain the information.

airmass,exptime

If the airmass and exposure time are not in the header nor can they be determined from other keywords in the header then these query parameters are used to request the airmass and exposure time. The values are updated in the image.

The following parameter is for the task to make queries.

answer

Interactive query parameter.

CURSOR KEYS

```
? Display help page
a Add a new band by marking the endpoints
d Delete band nearest the cursor in wavelength
r Redraw current plot
q Quit with current bandpass definitions
w Window plot (follow with '?' for help)
I Interrupt task immediately
```

```
:show Show current bandpass data
```

DESCRIPTION

Observations of standard stars are integrated over calibration bandpasses and written to an output file along with the associated calibration fluxes. The fluxes are obtained from tabulated standard star calibration files or a model flux distribution (currently just a blackbody) based on the magnitude and spectral type of the star. The output data is used by the task **sensfunc** to determine the detector sensitivity function and possibly the extinction. The spectra are required to be dispersion corrected. The input spectra may be in either "onedspec" or "echelle" format and may have many different observation apertures. The spectra may also be beam switched and use the a record number extension format.

The input spectra are specified by a list of names or root names if the record number extension format. In the latter case each name in the list has each of the specified record numbers appended. A subset of the

input spectra may be selected by their aperture numbers using the parameter *apertures*. The spectrum name, aperture number, and title are printed to the standard output. The airmass is required but if absent from the image header it may be computed from the observation header parameters and the latitude task parameter (normally obtained from the **observatory** task). If the airmass cannot be computed, due to missing keywords, then a query is made for the airmass. The airmass is then updated in the header.

The name of the standard star or blackbody curve is obtained by querying the user. If the parameter *samestar* is yes or beam switch mode is selected then all spectra are assumed to be of the same standard star and the query is made once. If the parameter is no then a query is made for each aperture. This allows each aperture to contain a different standard star. Note however that multiple observations with the same aperture number must be of the same standard star.

The standard star name is either the name of an actual standard star or of a blackbody calibration. The latter generally have a star name consisting of just the standard bandpass identifier. If the standard star name is not recognized a menu of the available standard stars in the calibration directory, the file "standards.men", is printed and then the query is repeated. Thus, to get a list you can type ?

The standard star names must map to a file containing tabulated calibration data. The calibration filename is formed from the star name with blanks, "+", and "-" removed, converted to lower case, and the extension ".dat" added. This name is appended to a calibration directory, so the directory name must have an appropriate directory delimiter such as "\$" or "/". Generally one of the system calibration directories is used but one may copy and modify or create new calibration files in a personal directory. For the current working directory the calibration directory is either null or "./".

The calibration files may include comment parameter information consisting of the comment character '#', a parameter name, and the parameter value. These elements are separated by whitespace. Any other comment where the first word does not match one of the allowed parameter names is ignored by the program. The parameter names are "type" identifying the type of calibration file, "units" identifying wavelength units, "band" identifying the band for magnitudes, and "weff" identifying the effective wavelength of the band.

There are two types of standard star calibration files as described below.

STANDARD STAR CALIBRATION FILES

This type of file is any file that does not contain the parameter "type" with a value of "blackbody". The only other parameter used by this type of calibration file is the "units" parameter for the wavelength units. If the units are not specified then the wavelengths default to Angstroms. All older calibration files will have no parameter information so they are interpreted as standard star calibration files with wavelengths in Angstroms.

The calibration files consist of lines with wavelengths, calibration magnitudes, and bandpass widths. The magnitudes are m_{AB} defined as

$$m_{AB}(\text{star}) = -2.5 * \log_{10} (f_{\nu}) - 48.60$$

where f_{ν} is in $\text{erg/cm}^2/\text{s/Hz}$. The m_{AB} calibration magnitudes are converted to absolute flux per unit frequency using the parameter *fnuzero* defined by

$$f_{\nu} = \text{fnuzero} * 10. ** (-0.4 * m_{AB})$$

Thus, *fnuzero* is the flux at m_{AB} of zero. The flux units are determined by this number. The default value was chosen such that Vega at 5556 Angstroms has an AB magnitude of 0.0336 and a $\text{ergs/cm}^2/\text{s/Hz}$. This is the same value that was used by all previous versions of this task.

BLACKBODY CALIBRATION FILES

This type of file has the comment parameter "type" with a "blackbody". It must also include the "band" and "weff" comment parameters. If no "units" comment parameter is given then the default units are Angstroms.

The rest of the file consists of lines with wavelengths, m_{AB} of a magnitude star (in that band magnitude system), and the bandpass widths. The m_{AB} are defined as described previously. Normally all the m_{AB} values will be the same though it is possible to adjust them to produce a departure from a pure blackbody flux distribution.

The actual m_{AB} calibration magnitudes for the star are obtained by the relation

$$m_{AB}(\text{star}) = \text{mag} + m_{AB}(m=0) - 2.5 * \log_{10} (B(\text{weff}, \text{teff})/B(w, \text{teff}))$$

where m is the magnitude of the star in the calibration band, $m_{AB}(m=0)$ is the calibration value in the calibration file representing the magnitude of a $m=0$ star (basically the m_{AB} of Vega), weff is the effective wavelength for the calibration file, and teff is the effective temperature of the star. The function $B(w,T)$ is the blackbody function in f_{ν} that provides the shape of the calibration. Note how the normalization is such that at weff the last term is zero and $m_{AB}(\text{star}) = m + m_{AB}(m=0)$.

The $m_{AB}(\text{star})$ computed using the calibration values and the blackbody function are then in the same units and form as for the standard star files. The conversion to f_{ν} and the remaining processing proceeds in the same way as for standard star calibration data.

The parameters $\backslash \text{mag}$ and teff are specified by the user for each star as described in the section BLACKBODY PARAMETERS. These parameters are queried by the task for each star (unless forced to a value on the command line).

The beam switch mode is selected with the *beam_switch* parameter. This mode requires that all apertures are of the same star, the header keyword OFLAG be present to identify object and sky spectra, and that the sequence of spectra specified are paired such that if an object spectrum is encountered first the next spectrum for that aperture (spectra from other apertures may appear in between) is a or the reverse. These restrictions are not fundamental but are made so that this mode behaves the same as with the previous version of this task. The sky spectrum is subtracted from the object spectrum and the result is then used in generating the observed intensities in the calibration bandpasses.

If the spectra have been extinction corrected (EX-FLAG = extinction correction is removed. The specified extinction file is used for this operation and so must be the same as that used when the extinction correction was made. The airmass is also required in this step and, if needed to compute the airmass, the observatory specified in the image or observatory parameter is used. The treatment of extinction in this task is subtle. The aim of this task is to produce observed integrated instrumental intensities without extinction correction. Thus, the extinction correction is removed from extinction corrected spectra. However, a correction is made for an extinction gradient across the bandpasses. This is done by applying an extinction correction, integrating across the bandpass, and then correcting the integrated intensity for the extinction at the center of the bandpass. An alternative way to look at this is that the integral is weighted by the ratio of the extinction correction at each pixel to the extinction correction at the center of the bandpass. This correction or weighting is why the extinction file and latitude are parameters in this task even though for nonextinction corrected spectra they appear not to be needed.

The observed instrumental intensities are integrated within a bandpasses by summing the pixels using partial pixels at the bandpass edges. Initial bandpasses are defined in one of two ways. A set of evenly spaced bandpasses of constant width covering the range of the input spectrum may be specified using the parameters *bandwidth* and *bandsep* in the same units as the spectrum dispersion. If these parameters have the value INDEF then the bandpasses from the calibration file which are entirely within the spectrum are

selected. Generally these bandpasses are the actual measured bandpasses though one is free to make calibration files using estimated points. The calibration bandpasses are preferable because they have been directly measured and they have been placed to avoid troubles with spectral lines. However, when the coverage or resolution is such that these bandpasses do not allow a good determination of the instrumental response the evenly spaced bandpasses may be needed. The calibration fluxes are linearly interpolated (or extrapolated) from the calibration data points to the defined bandpasses.

Each spectrum adds a line to the output file containing the spectrum image name, the sky spectrum image name if beam switching, the aperture or beam number, the number of points in the spectrum, the exposure time, airmass, wavelength range, and title. If the airmass is not found in the image header it is computed using the latitude parameter and observation information from the header. If the airmass cannot be computed, due to missing keywords, then a query is made for the airmass.

Following the spectrum information, calibration data is added for each bandpass. The bandpass wavelength, absolute flux (per Angstrom), bandpass width, and observed instrumental intensity in the bandpass are added to the output file. As discussed above, the observed intensity does not include an extinction term but does apply a small correction or weighting for the variation of the extinction across the bandpass.

The setting and editing of the bandpasses may be performed interactively if the *interact* flag is set. In this case the user is queried for each spectrum. The answers to this query may be "no" or "yes" to skip editing or edit the bandpasses for this spectrum, "NO" or "YES" to skip or not skip editing all spectra of the same aperture with no further queries for this aperture, and "NO!" or "YES!" to skip editing or edit all spectra with no further queries.

When editing the bandpasses a graph of the spectrum is made with the bandpasses plotted at the computed intensity per pixel. The cursor and colon commands available are summarized in the section CURSOR KEYS. Basically bandpasses may be added or deleted and the current bandpass data may be examined. Additional keys allow the usual windowing and cursor mode operations. When satisfied with the bandpasses exit with 'q'. The edited bandpasses for that aperture remain in effect until changed again by the user. Thus if there are many spectra from the same aperture one may reply with "NO" to queries for the next spectra to accept the current bandpasses for all other spectra of the same aperture.

BLACKBODY PARAMETERS

When a blackbody calibration is selected (the calibration file selected by the *star_name* parameter has "# type blackbody") there are two quantities needed to scale the blackbody ~~the~~ observation. These are the magnitude of the star in the same band as the observation and the effective temperature. The magnitude is used for the flux scaling and the effective temperature for the shape of the flux distribution. The values are obtained or derived from the user specified parameters *mag*, *magband*, and *teff*. This section describes how the values are derived from other parameters using the data file "params.dat" in the calibration directory.

The effective temperature in degrees Kelvin may be specified directly or it may be derived from a spectral type for the star. In the latter case the file "params.dat" is searched for the effective temperature. The file consists of lines with the first value being the spectral type and the second the effective temperature. Other columns are described later. The spectral type can be any string without whitespace that matches what is in the file. However, the program finds the last spectral type that matches the first two characters when there is no complete match. This scheme is intended for the case where the spectral types are of the form A0I, A0III, or A0V, where A can be any spectral type letter OBAFGKM, the single digit subtype is between 0 and 9, and the luminosity class is one of I, III, or V. The two character match selects the last spectral type independent of the luminosity class. The standard file "onedstds\$blackbody/params.dat" uses these spectral type identifiers with the dwarf class acting as the default.

The magnitude band is specified along with the input magnitude. If band is the same as the calibration band given in the calibration file then no further transformation is required. However if the magnitude is

specified in a different band, a conversion is performed using information from the "params.dat" file based on the spectral type of the star.

When an effective temperature is specified rather than the nearest tabulated temperature for the spectral types that have "V" as the third character is used. For the standard spectral type designations this means that when an effective temperature is specified the dwarf spectral type is used for the magnitude transformation.

As mentioned previously, the "params.dat" data file has additional columns following the spectral type and effective temperature. These columns are relative magnitudes in various bands. The standard file has V magnitudes of zero so in this case the columns are also the X-V colors (where X is appropriate magnitude). Given the spectral type the relative magnitudes for the calibration band, m_1, and the input magnitude band, m_2, are found and the calibration magnitude for the star is given by

$$m_{\text{calibration}} = m_{\text{input}} + m_1 - m_2$$

If one of the magnitudes is missing, given as "INDEF" because the transformation is not available for the spectral type, the last spectral type matching the first two characters which does specify the two magnitudes will be used. For example if there is no information for a B3III star for a M-J color then the spectral type B3V might be used.

In order for the program to determine the bands for each column in the data file there must be a comment before the data with the column names. It must begin with "# Type Teff" and then be followed by the same band identifiers used in the blackbody calibration files and as specified by the *magband* parameter. Any amount whitespace (space or tab) is used to separate the various fields in the comment and in the fields of the table. For example the file might have the comment

```
# Type      Teff      V      J      H      K      L      Lprime      M
```

identifying the third column of the file as the V magnitude and the ninth file as the M magnitude.

EXAMPLES

1. To compile observations of three standard stars using a switched instrument like the IIDS:

```
cl> standard.recformat=yes
cl> standard nite1 1001-1008 std beam_switch+ interact-
[nite1.1001][0]: HZ 44 - Night 1
[nite1.1004][0]: HZ 44 - Night 1
[nite1.1005][0]: HZ 44 - Night 1
[nite1.1008][0]: HZ 44 - Night 1
Star name in calibration list: hz 44
cl> standard nite1 1009-1016 std beam_switch+ interact-
...
cl> standard nite1 1017-1024 std beam_switch+ interact-
...
```

This will create a file "std" which will contain sensitivity measurements from the beam-switched observations of the three standard stars given. Note that **standard** is run separately for each standard star.

The spectra will be from the images: nite1.1001, nite.1002 ... nite1.1024, and the default calibration file, "onedstds\$irscal.dat" will be used.

2. For echelle spectra all apertures, the orders, are of the same star and so the samestar parameter is set. Usually the resolution is much higher than the calibration data so in order to get sufficient coverage bandpasses must be interpolated from the calibration data. Therefore the evenly spaced bandpasses are used.

```
cl> standard.recformat=no
cl> standard.samestar=yes
```

```

cl> standard ech001.ec std bandwidth=10 bandsep=15
[ech001.ec][0]: Feige 110
Star name in calibration list: feige 110
[ech001.ec][0]: Edit bandpasses? (no|yes|NO|YES|NO!|YES!): yes
[ech001.ec][1]: Edit bandpasses? (no|yes|NO|YES|NO!|YES!): yes
[ech001.ec][2]: Edit bandpasses? (no|yes|NO|YES|NO!|YES!): NO!

```

3. To use a blackbody infrared calibration where the V magnitude of the star is known.

```

cl> standard std1.ms std caldir=onedstds$blackbody/
std1.ms(1): Standard Star
Star name in calibration list: J
Magnitude of star: 10.3
Magnitude type (|V|J|H|K|L|Lprime|M|): V
Effective temperature or spectral type: B3III
WARNING: Effective temperature for B3III not found - using B3V
Blackbody: V = 10.30, J = 10.32, Teff = 19000
std1[1]: Edit bandpasses? (no|yes|NO|YES|NO!|YES!) (yes):

```

Note the warning message and the confirmation information.

REVISIONS

STANDARD V2.10.4

The calibration files can be defined to compute blackbody values.

STANDARD V2.10.3

A query for the airmass and exposure time is now made if the information is not in the header and cannot be computed from other header keywords.

STANDARD V2.10

Giving an unrecognized standard star name will page a stars available in the calibration directory and then repeat the query.

SEE ALSO

observatory, lcalib, sensfunc